Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/02/2021
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Gravity Scaling of Pool Boiling Heat Transfer: Numerical Simulations and Validation with MABE and NPBX		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/18/2021 per NSSC information (Ed., 7/27/2020) NOTE: Extended to 8/18/2020 per NSSC information (Ed., 11/12/19) NOTE: Extended to 8/15/2019 per NSSC information (Ed., 10/3/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kim, Jungho Ph.D. (University of Maryland) Balaras, Elias Ph.D. (The George Washington University)		
Grant/Contract No.:	NNX16AQ77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The key challenge in developing two-phase thermal systems for space environments is the development of a heat transfer database and reliable models for boiling in variable gravity environments from which the performance of two-phase heat exchangers can be predicted. The Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. Earth gravity data obtained prior to launch served as the reference data. The ability of a model developed from parabolic aircraft work and ISS data to predict how boiling was altered under a range of gravity levels was tested and found to predict the ISS data within 20%. The focus of NPBX was the investigation of how single and multiple bubbles (up to 5) grew and merged in microgravity. Heat fluxes for steady state nucleate boiling and critical heat flux were found to be much lower than those obtained under Earth gravity. It was found that heater size, chamber geometry, and bulk fluid conditions must be accounted for in nucleate boiling heat transfer correlations under microgravity conditions. In the proposed work, high fidelity numerical simulations of the boiling process will be used to glean further insights into the boiling process. The data from MABE and NPBX will be used to validate the simulations as they provide a wide range conditions (heater sizes, subcoolings, and superheats including CHF (critical heat flux)) at which to test the numerical results. NPBX will serve as validation for simulations with a few nucleation sites in microgravity. Simulations of boiling in Earth gravity will be made for the MABE heater and compared with the MABE Earth gravity results, after which additional simulations will be performed as gravity is slowly decreased. The results will be compared with the MABE microgravity results and the model. Co-Investigators for the proposed work are Prof. Elias Balaras (George Washington University) and Prof. Jungho Kim (U. of Maryland, PI for MABE).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reliable, physics based direct numerical simulation techniques that are also numerically efficient will be used to interrogate the fundamental physics of the boiling process under low-gravity and Earth-gravity conditions. Such high accuracy modeling of the boiling process will help develop effective heat transfer methodologies for the cooling of various engineering equipment including microprocessors.
Task Progress:	FINAL REPORTING DECEMBER 2021 Nucleate pool boiling simulations were carried out to study the effects of gravity on the dynamics of bubble formation, bubble departure, and the mutual bubble interactions using high fidelity numerical simulations. Research supported by this project has led to the development and coding, as well as verification and validation, of a software for scientific computing, called "Numerical Solver" in this report, to simulate the physical behavior of boiling heat transfer. The solver is fully three-dimensional and uses state-of-the-art computational methods for modeling the physics of thermo-fluid processes involved in boiling heat transfer in both low and Earth normal gravity environments. The solver is capable of utilizing parallel processing on the order of thousands of processors using NASA Advanced Super Computing facilities. The simulations have revealed that the dynamic contact line model is integral in modeling inertial effects in pool boiling heat transfer, especially in low gravity conditions. Absence of inertial effects when using a static contact line model leads to premature departure. This may not be evident in Earth gravity, but in low gravity it causes non-physical bubble detachment, which leads to higher-than-normal heat fluxes. Available methods in literature use only a static contact angle for boiling simulations since their analysis is limited to Earth gravity where overall physics is less sensitive to inertial effects because buoyancy dominates the flow. The simulations identified the need for a dynamic contact line model because accounting for inertial effects is critical in low gravity due to the significance of surface tension. This dynamic contact line model is coupled with a smeared microlayer model to account for sub-grid scale heat transfer. Simulations carried out using the developed solver have identified dominant trends associated with the flow and thermal behavior at different gravity levels, using fixed values of wall superheat and liquid subcooling. This was accomplished by measuring the contributions to heat flux associated with turbulent flow resulting from the dynamics of bubble interaction. The time and space averaged flow quantities helped us capture the dominant trends within the instantaneous and local turbulent fluctuations. The time averaged fluctuations of velocity and temperature indicate that the wall heat flux conditioned by the wetted surface area – the liquid heat flux – were used to describe high heat flux hot spots. Such patterns underscore the highly transient nature of bubble formation. Despite substantial differences in the dynamics of large-scale bubbles at different gravity levels, the average heat flux in the liquid region was found to be dominated by the sliding and merger type events associated with small-scale bubbles over a wide range of gravity levels. These observations reveal the significance of small-scale dynamics in affecting the heat transfer, which was previously thought to be dominated by large-scale bubbles, particularly at low gravity levels. The fluctuations of velocity and temperature were measured in order to determine the turbulent heat flux. While the maximum value of average turbulent fluctuations occurs away from the wall for Earth normal gravity, the near-wall peak was found to be only slightly lower and similar in magnitude to that in the case of low gravity. The Q-criterion was used for determining the production of vorticity related to bubble dynamics. The significance of near-wall, small-scale bubble dynamics was reflected in the transition of vorticity structures from hairpin type vortices in the near-wall region to vortex ring patterns away from the wall. The behavior of associated turbulence intensity indicates how the peak in turbulence intensity shifts with distance from the wall for different gravity levels. The correlation of bubble dynamics with vorticity generation and turbulence intensity shows that the near-wall flow behavior could be similar even at very different gravity levels. This is further supported by the near-wall distribution of turbulent heat flux. We further filtered the turbulent heat flux using a criterion based on the proximity of nucleation sites that provides information about the contribution of bubble interaction to the total turbulent heat flux. These results have been reported into two full-length manuscripts that have appeared in the International Journal of Multiphase Flow and the International Journal of Heat and Mass Transfer. (Ed. Note: see Cumulative Bibliography). ANNUAL REPORTING AUGUST 2021 Nucleate pool boiling simulations were carried out to study the effects of gravity on the dynamics of bubble formation, bubble departure, and the mutual bubble interactions. We focused on identifying the dominant trends associated with the flow and thermal behavior at different gravity levels using fixed values of wall superheat and liquid subcooling.

	This was accomplished by measuring the contributions to heat flux associated with turbulent flow resulting from the dynamics of bubble interaction. The time and space averaged flow quantities helped us capture the dominant trends within the instantaneous and local turbulent fluctuations. The time averaged fluctuations of velocity and temperature indicate that the wall heat flux conditioned by the wetted surface area - the liquid heat flux – were used to describe high heat flux hot spots. Such patterns underscore the highly transient nature of bubble formation. Despite substantial differences in the dynamics of large-scale bubbles at different gravity levels, the average heat flux in the liquid region was found to be dominated by the sliding and merger type events associated with small scale bubbles over a wide range of gravity levels. This significance of near-wall, small scale bubble dynamics was reflected in the transition of vorticity structures from hairpin type vortices in the near-wall region to vortex ring patterns away from the wall. While the maximum value of average turbulent fluctuations occurs away from the wall for Earth normal gravity, the near-wall peak was found to be only slightly lower and similar in magnitude to that in the case of low gravity. These observations reveal the significance of small-scale dynamics in affecting the heat transfer which was previously thought to be dominated by large scale bubbles, particularly at low gravity levels. These results have been incorporated into a manuscript that is currently under review at International Journal of Heat and Mass Transfer. The first round of reviews was encouraging and we have submitted a revision.
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Articles in Peer-reviewed Journals	Dhruv A, Balaras E, Riaz A, Kim J. "An investigation of the gravity effects on pool boiling heat transfer via high-fidelity simulations." International Journal of Heat and Mass Transfer. 2021 Dec;180:121826. https://doi.org/10.1016/j.ijheatmasstransfer.2021.121826 , Dec-2021